

Common Checked Exceptions in Java with Examples

In Java, checked exceptions are exceptions that must either be caught in a try-catch block or declared in a method's throws clause. Below are common checked exceptions, their causes, and example code snippets that demonstrate how they might occur.

IOException

Cause: Errors during input/output operations (file, network).

Example:

```
try {  
    FileReader fr = new FileReader("test.txt");  
} catch (IOException e) {  
    e.printStackTrace();  
}
```

FileNotFoundException

Cause: File not found at specified path.

Example:

```
try {  
    FileReader fr = new FileReader("missing.txt");  
} catch (FileNotFoundException e) {  
    e.printStackTrace();  
}
```

EOFException

Cause: Unexpected end of file reached.

Example:

```
try {  
    ObjectInputStream ois = new ObjectInputStream(new  
        FileInputStream("data.dat"));  
    ois.readObject();  
} catch (EOFException e) {  
    e.printStackTrace();  
}
```

SQLException

Cause: Database access or invalid SQL query.

Example:

```
try {  
    Connection con =  
        DriverManager.getConnection("jdbc:mysql://localhost/db", "user",  
            "pass");  
    Statement st = con.createStatement();  
    st.executeQuery("INVALID SQL");  
} catch (SQLException e) {  
    e.printStackTrace();  
}
```

ClassNotFoundException

Cause: Class not found when loading dynamically.

Example:

```
try {  
    Class.forName("com.example.UnknownClass");  
} catch (ClassNotFoundException e) {  
    e.printStackTrace();  
}
```

InterruptedException

Cause: Thread interrupted during sleep/wait.

Example:

```
try {  
    Thread.sleep(1000);  
} catch (InterruptedException e) {  
    e.printStackTrace();  
}
```

InvocationTargetException

Cause: Exception thrown by invoked method via reflection.

Example:

```
try {  
    Method m = String.class.getMethod("charAt", int.class);  
    m.invoke("test", -1);  
} catch (InvocationTargetException e) {
```

```
e.printStackTrace();  
}
```

NoSuchMethodException

Cause: Method not found using reflection.

Example:

```
try {  
    Method m = String.class.getMethod("nonExistent");  
} catch (NoSuchMethodException e) {  
    e.printStackTrace();  
}
```

InstantiationException

Cause: Attempt to instantiate abstract class/interface.

Example:

```
try {  
    Class<?> clazz = AbstractList.class;  
    Object obj = clazz.newInstance();  
} catch (InstantiationException e) {  
    e.printStackTrace();  
}
```

ParseException

Cause: Invalid parsing of text to date/number.

Example:

```
try {  
    SimpleDateFormat sdf = new SimpleDateFormat("dd/MM/yyyy");  
    Date d = sdf.parse("invalid-date");  
} catch (ParseException e) {  
    e.printStackTrace();  
}
```

GeneralSecurityException

Cause: Cryptography/security-related issues.

Example:

```
try {  
    KeyGenerator.getInstance("InvalidAlgo");  
} catch (GeneralSecurityException e) {  
    e.printStackTrace();  
}
```

URISyntaxException

Cause: String cannot be parsed as URI.

Example:

```
try {  
    URI uri = new URI("ht!tp://bad uri");  
} catch (URISyntaxException e) {  
    e.printStackTrace();  
}
```
